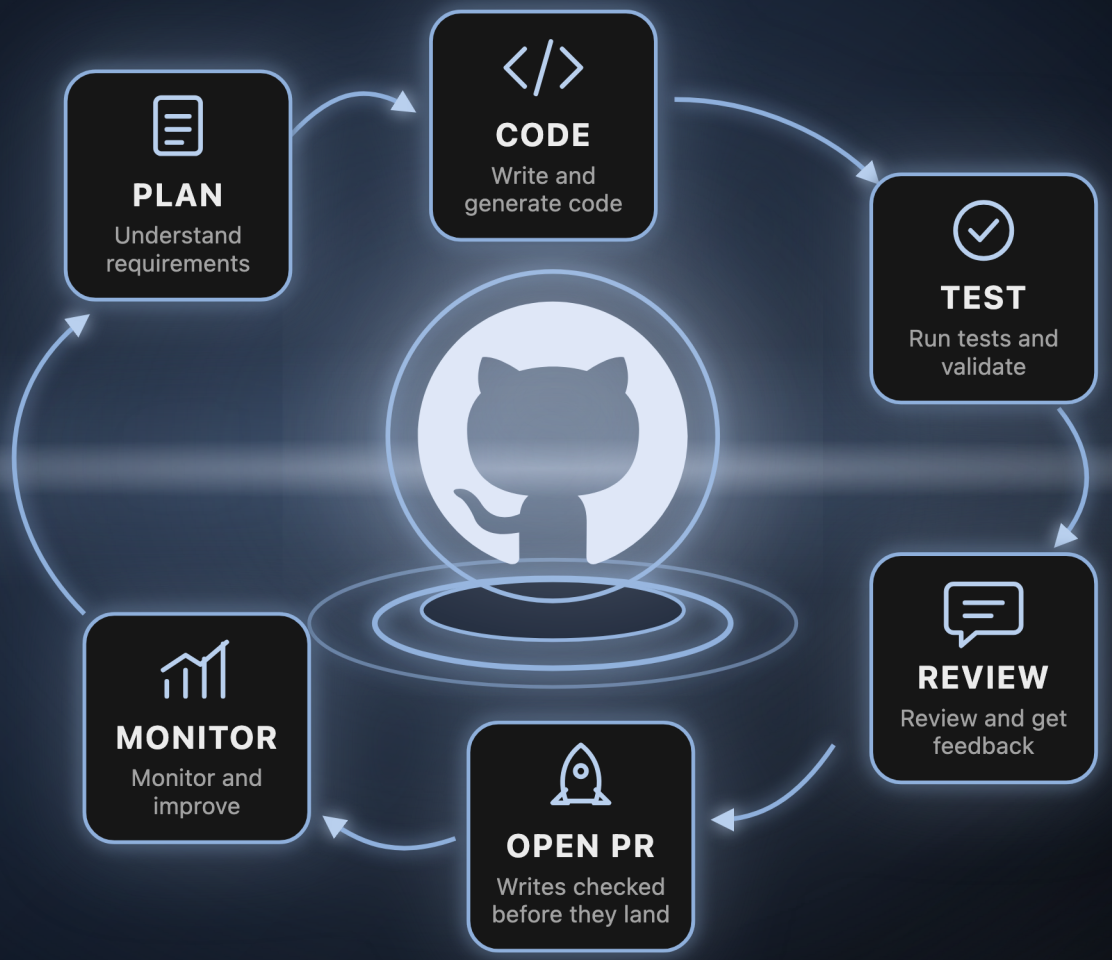
GitHub

Repos that keep themselves up to date

GitHub Agentic Workflows: from AI that suggests to AI that does

Donal Moloney TODO: talk date and audience



Somewhere in our docs, a page is wrong right now, and nobody knows which one

56.5%

of 12,374 developers say **outdated information** gets in the way of the context they need at work.

Nobody chose to let that page go stale: keeping it current is nobody's job

Three people could have fixed it. Each had a better reason not to.

THE AUTHOR

Shipped the change that made the page wrong, and moved to the next ticket. The code was the deliverable. The page was not.

THE REVIEWER

Is reading a diff against the clock. The stale sentence is in another file, in another folder, and nothing fails without it.

THE NEXT READER

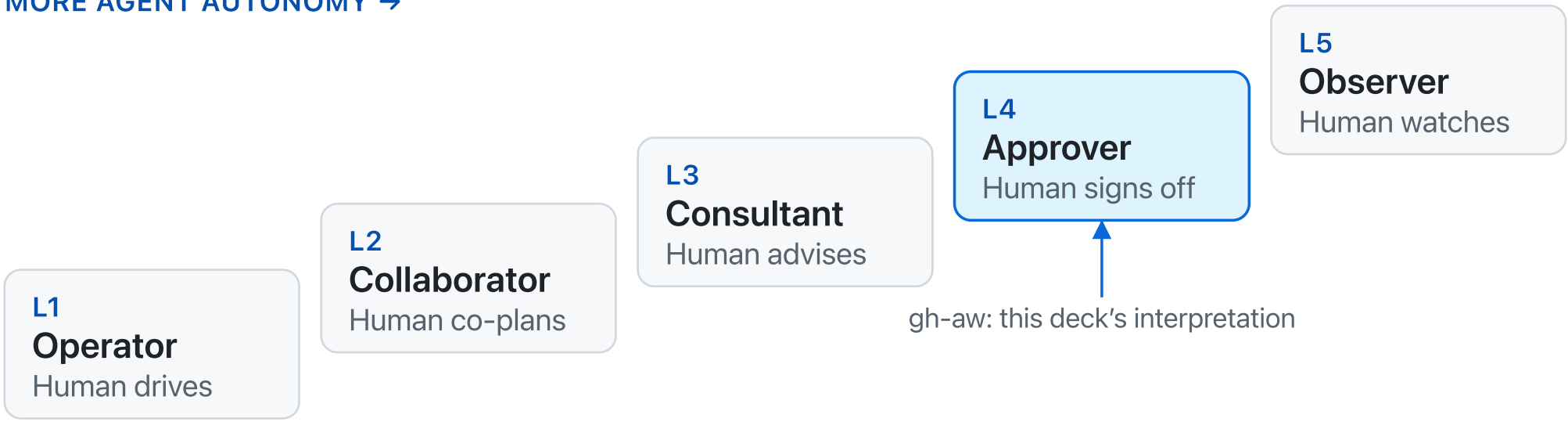
Hits the wrong page, works around it, and has no reason to adopt a file they have never edited.

So more discipline will not fix this. The update is **always worth doing and never worth doing now** — and a job that never wins the week needs an owner with nothing else competing for the time.

This slide is the deck's own argument, not a cited finding.

We ship more code than ever, and we are choosing how much of each change to hand to an agent

MORE AGENT AUTONOMY →



- **43.2 million** pull requests merged per month, up 23% year over year.
- **Nearly 1 billion** commits pushed in 2025, up 25.1%.

Measured: GitHub Octoverse 2025. Ladder: Feng et al., arXiv 2506.12469, illustrative; the L4 placement is this deck's interpretation.

I built something similar before, and it showed me what was missing

What I built	What worked	What was missing
TODO	TODO	TODO

TODO: Add screenshot and the specific lesson from the earlier project.

An agentic workflow is a GitHub Action you brief instead of script

A GitHub Action is a job your repository runs by itself whenever something happens in it.

HIRING A TEAMMATE		IN THE WORKFLOW FILE
Job description what they are asked to do	→	**Markdown body** plain-English instructions
Shift when they work	→	**`on:`** when the workflow runs
Access badge what they may touch	→	**`permissions, tools, network`** what it can read and reach
Outbox you check what they hand back	→	**`safe-outputs`** approved writes, reviewed

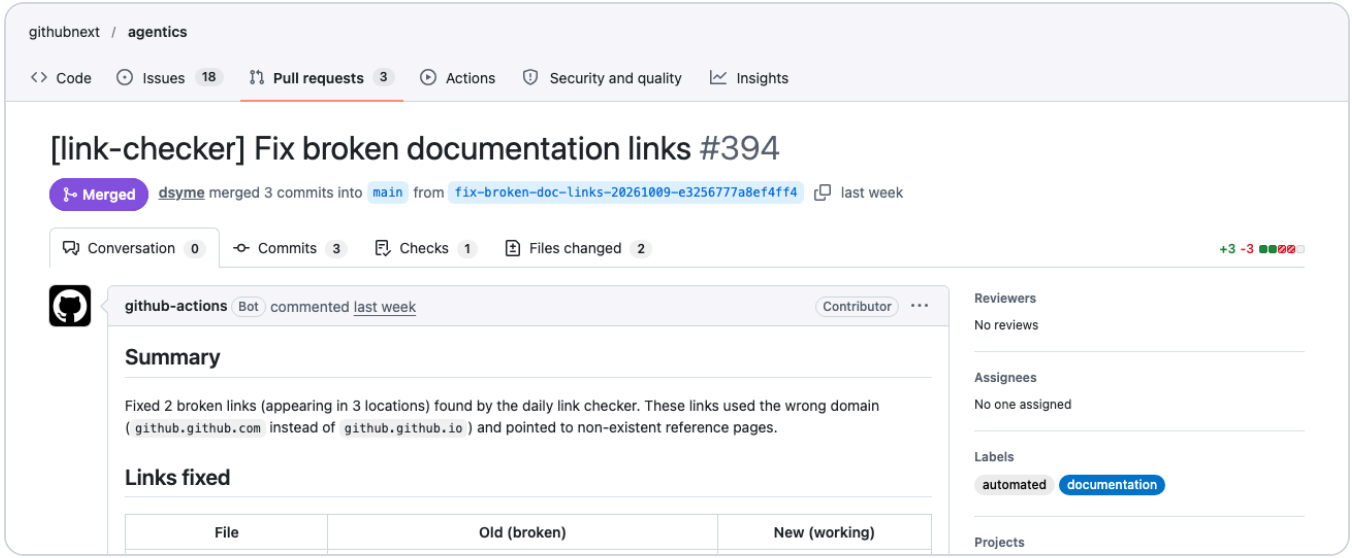
Classic Action: you write the steps. Agentic workflow: you write the brief, the agent plans the steps.

The teammate mapping is this deck's model, not GitHub terminology.

GitHub reports that 57 of 59 pull requests from its docs workflow were merged

57 / 59

A 96% merge rate for its Daily Documentation Updater.



A different workflow in the same repo — the link checker, not the docs updater.

Number: gh-aw blog, 13 Jan 2026; reported by GitHub, small sample — merged is not the same as correct. Screenshot: real merged PR `githubnext/agentics#394`, captured 18 Oct 2026.

It will get things wrong, so it is built to be checked, not trusted

You would review a new teammate's first pull request. This one cannot skip that step.

WRONG EDITS

The workflow file decides. `draft: true` opens a draft the agent cannot override — but it is set per workflow, not guaranteed by the platform.

Never merged automatically: GitHub Blog guidance, not a hard limit.

PROMPT INJECTION

The agent is read-only and fenced in. The next slide shows how.

COST

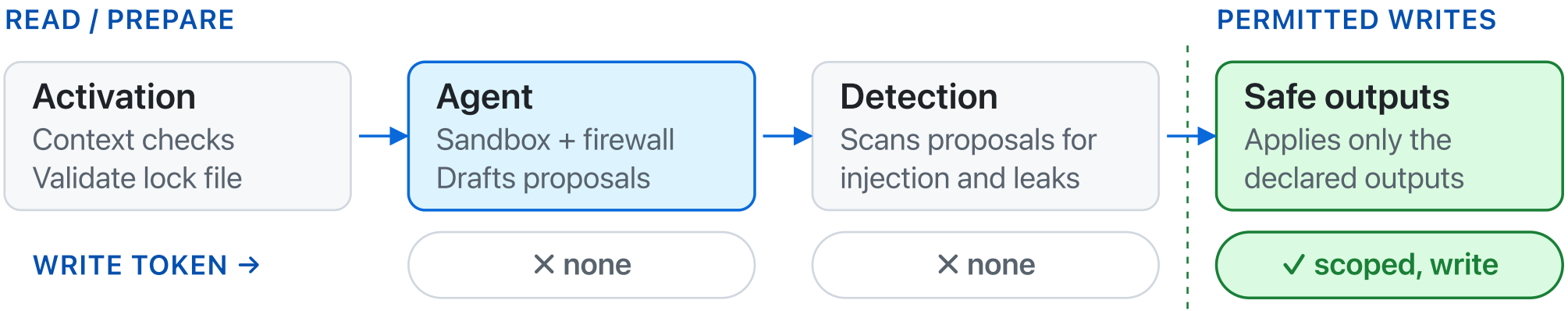
Bounded per run by default, but a month's bill depends on how often it starts. We come back to that before the ask.

NOT A FIT

It does not replace build, test or release pipelines.

gh-aw docs (safe outputs, cost management); GitHub Blog.

The agent never holds a write token: a separate, checked job does the writing



Only the proposed output is handed on from job to job

The teammate drafts the letter; someone else holds the key to the postbox.

- Detection scans every proposed output before anything is written. A failed check defaults to a caution notice, not a hard block (continue-on-error is true).

gh-aw architecture documentation; four of its six jobs shown.

Thirteen lines say when it runs, what it can read, and the one thing it can send

```yaml
on:
  push:
    branches: [main]
permissions: read-all
network: defaults
safe-outputs:
  create-pull-request:
    draft: true
tools:
  github:
    toolsets: [all]
  web-fetch:
  bash: true
```

- **Shift:** every push to `main`
- **Access badge:** read-only; network limited to basic infrastructure
- **Outbox:** one draft pull request
- **Tools:** broad **reads** — `bash`, `web-fetch`, every GitHub read toolset.

> Yes, that is a lot of access. All of it runs **inside the read-only, firewalled job** from the previous slide, and the outbox is still the only way anything leaves.

Frontmatter trimmed from `workflows/update-docs.md` in githubnext/agentics.

The library is already written and documented; running it is the one step left

Already written	Still unproven
98 workflow briefs in `workflows/`, one file each	**0** have ever run inside GitHub Actions
98 matching docs pages, one per workflow	**4 of 98** compiled locally; the other 94 are briefs
6 shared policy and prompt modules, imported rather than copied	No release tagged, and the sandbox repo does not exist

TODO: the one thing building this taught me, and whether the repo goes public before the talk.

`DonalMoloney/Agentic-Workflows-Lib`, private. File counts verified via the GitHub API, 17 Oct 2026. The compile result and the "not yet run" column are the repository's own `agents.md`, a self-report, not a CI result.

A month costs whatever we let it cost, and the run count is a line in a file we review

The per-run spend is capped out of the box. What a month costs is a decision about how often it is allowed to start.

WE CHOOSE

runs per month

A schedule is a known number: daily is about thirty. Every push to `main`, as the example file used, is not a number anyone controls.

×

CAPPED BY DEFAULT

cost per run

1,000 AI Credits, about $10. The model is also ours to pick.

=

WHAT WE COMMIT

the monthly ceiling

TODO: our cap, our expected run count, and the ceiling I am asking for.

There is no surprise bill here. Both numbers are **settings in the workflow file**, reviewed in a pull request like any other code.

gh-aw cost management docs; 1 AI Credit = $0.01, which the docs call best-effort. The daily cap is left off on purpose: two docs pages disagree on its default.

Pick one stale page, and let an agent keep it current for a month

HYPOTHESIS

An agent can keep one page current at a quality we would merge.

METHOD

One page, one repository, four weeks. Staged first (`staged: true` runs it for real and writes nothing), then draft pull requests only, every one reviewed.

MEASURE

Two numbers: the share of its pull requests we merge without edits, and whether the page is actually right at the end of the month.

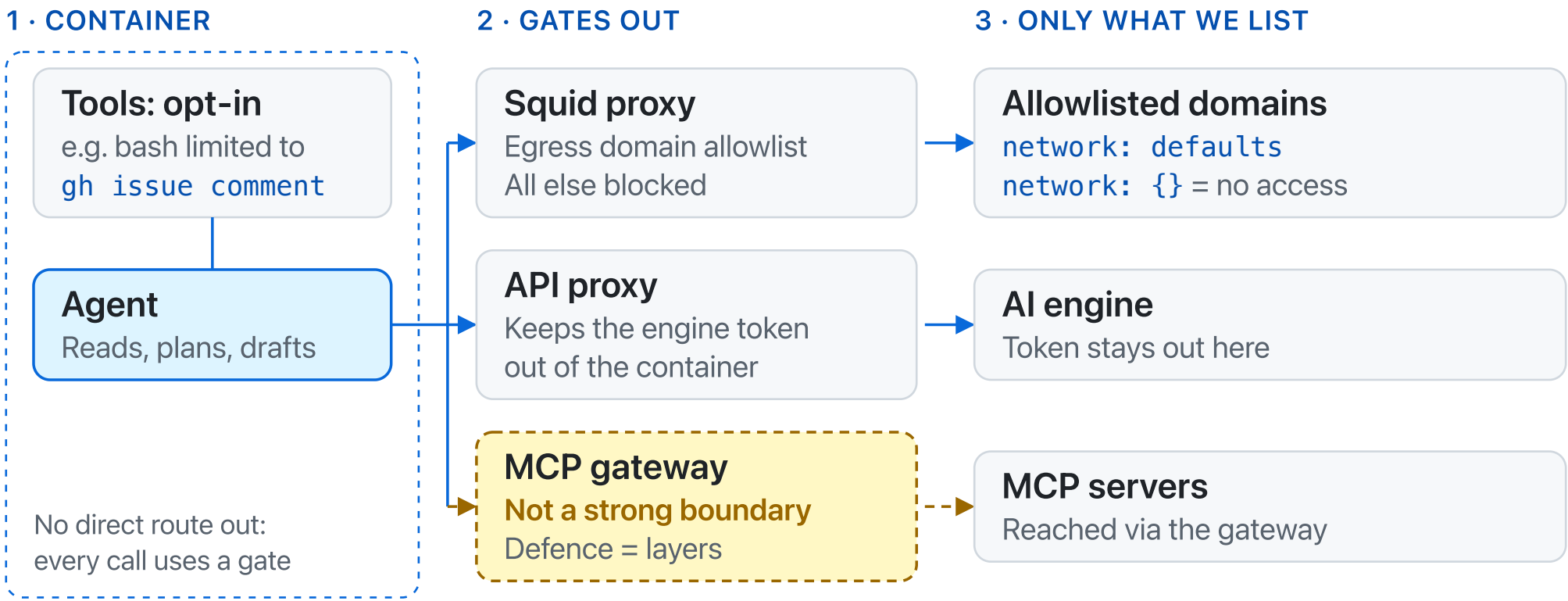
Merge rate alone measures agreement, not freshness. GitHub's 96% is a number to test, not to assume.

DECISION RULE

TODO: the result that means keep, the result that means stop.

What I need today: TODO: the pilot repository, its owner, and a budget.

A misled agent is boxed in by layers, not by any single wall



gh-aw architecture and network reference. The MCP gateway caveat is the documentation's own wording.

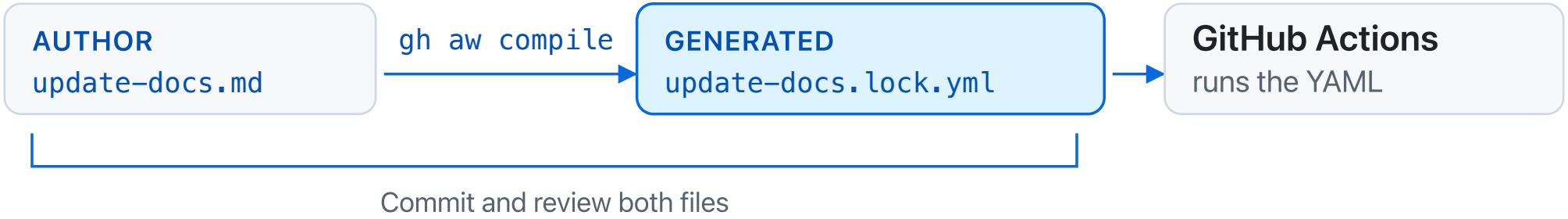
You debug it like any Action: every run leaves logs you can audit

1. **Review the change:** the `.md` workflow and generated `.lock.yml` in the pull request.
2. **Watch the run:** the Activation, Agent, Detection and Safe outputs jobs — four of the six.
3. **Dig in:** `gh aw logs <workflow>` across runs, `gh aw audit <run-id>` for one run.

> “Treat the workflow Markdown as code. Review changes, keep it small, and evolve it intentionally.”
>
> GitHub Blog

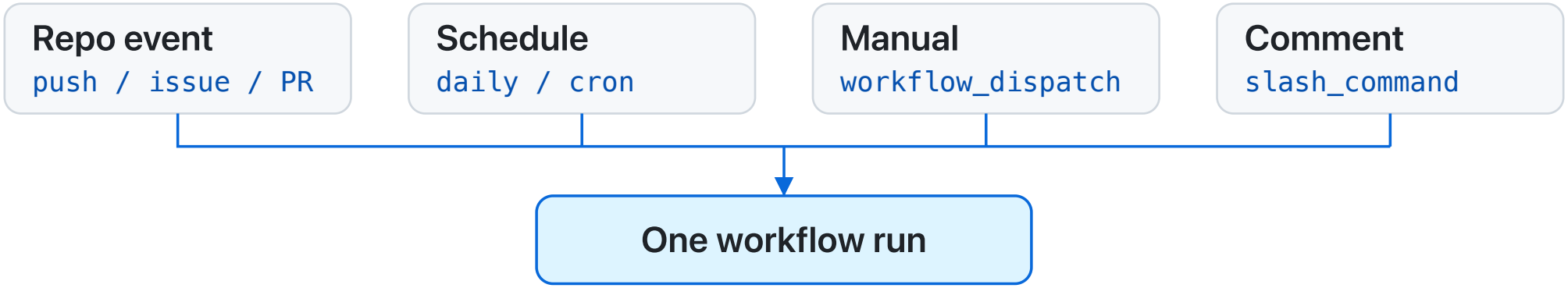
The instructions are plain English, and one command locks them into a GitHub Action

```
---
(on:, permissions, network, safe-outputs)
---
Update the docs to match the code that changed, and open a draft PR.
```



Body paraphrased; the real file's instructions are longer.

You start a workflow four ways: a repo event, a clock, a button, or a comment



Try it safely first

- `gh aw trial` runs a workflow in a temporary private repository (experimental).
- `staged: true` under `safe-outputs:` previews every output and writes nothing.

Try it yourself: my examples and the official docs

- TODO: My sites and example repository URLs
- `https://github.github.com/gh-aw` — the docs, and `/gallery` for ready-made workflows
- `https://github.com/githubnext/agentics` — where this deck's example `update-docs.md` comes from
- `https://github.blog/changelog/2026-06-11-github-agentic-workflows-is-now-in-public-preview`

Status and sources

- Fact-checked 17 Oct 2026 against the gh-aw documentation, GitHub changelog, Stack Overflow Developer Survey 2026, Octoverse 2025, GitHub Next, and the GitHub Blog sources cited on each slide.
- Status: **public preview since 11 Jun 2026** (GitHub changelog). The gh-aw docs site still links a technical-preview deck; treat that as a leftover.
- Left off on purpose: the daily AI Credit cap (cost-management page says 5,000 per workflow per day by default; frontmatter page says `max-daily-ai-credits` is disabled by default), and “always blocks” for detection (`continue-on-error` defaults to `true`, turning detection warnings/failures into a caution notice).